

Chapter 5 機構運動分析——解析法

5.2 若曲柄滑塊四連桿桿長及角度關係如圖 5.15 中所示，若以 O_2 為座標原點，請問 B 點座標為多少？另當 $\theta_2 = 120^\circ$ 時，請問 θ_3 為幾度？ B 點座標為多少？

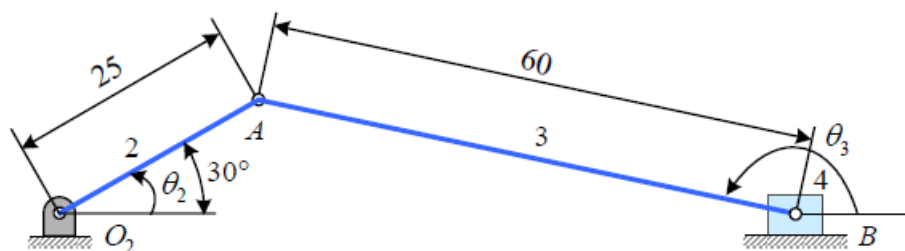


圖 5.15

解：

向量迴路法，參考(5.32)之向量迴路方程式，並依(5.34)~(5.39)方式，將 $\theta_2 = 30^\circ$ 代入求得 $\theta_3 = 168^\circ$ 及 $r_1 = 80.33$ ，則 B 點座標為 $(80.33, 0)$ ；將 $\theta_2 = 120^\circ$ 代入求得 $\theta_3 = 158.8^\circ$ 及 $r_1 = 43.45$ ，則 B 點座標為 $(43.45, 0)$ 。

5.6 表 5.1 定義了六組銷接頭四連桿組各桿件長度及輸入角 θ_2 的數據，連桿組合方式及變數名稱標示如圖 5.18，請以向量迴路法（複數向量法）求出 θ_3 及 θ_4 （即找出 $(\theta_{31}, \theta_{41})$ 與 $(\theta_{32}, \theta_{42})$ ）的可能角度。

表 5.1 銷接頭四連桿組各桿件長度及輸入角

列	連桿 1	連桿 2	連桿 3	連桿 4	θ_2
1	6	2	7	9	45
2	20	10	10	10	50
3	9	7	11	6	120
4	3	10	6	8	40
5	9	7	11	8	50
6	4	6	10	7	85

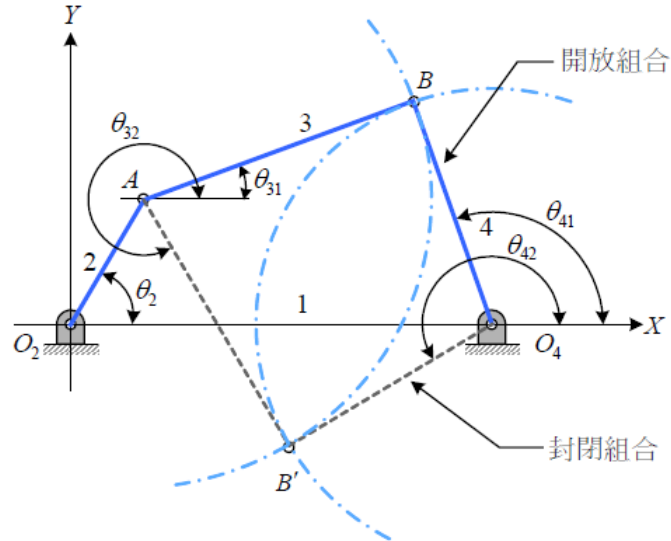


圖 5.18

解：

以列 1 為例，其位置向量方程式如下：

$$\vec{R}_A + \vec{R}_{BA} - \vec{R}_{BO_4} - \vec{R}_{O_4} = 0$$

或以複數向量表示為 $r_2 e^{j\theta_2} + r_3 e^{j\theta_3} - r_4 e^{j\theta_4} - r_1 e^{j\theta_1} = 0$

代入列 1 參數得

$$2(\cos 45^\circ + j \sin 45^\circ) + 7(\cos \theta_3 + j \sin \theta_3) - 9(\cos \theta_4 + j \sin \theta_4) - 6(\cos 0^\circ + j \sin 0^\circ) = 0$$

參考(5.14)~(5.31)方式分解成實部及虛部計算可得 (θ_3, θ_4) 為 $(81^\circ, 112^\circ)$

或 $(-115^\circ, -147^\circ)$

5.9 在圖 5.21 若 $\omega_2 = 150 \text{ r/min}$ ，試利用向量迴路法 (複數向量法) 求出向量

$\bar{V}_C$ 及 $\omega_4 = ?$

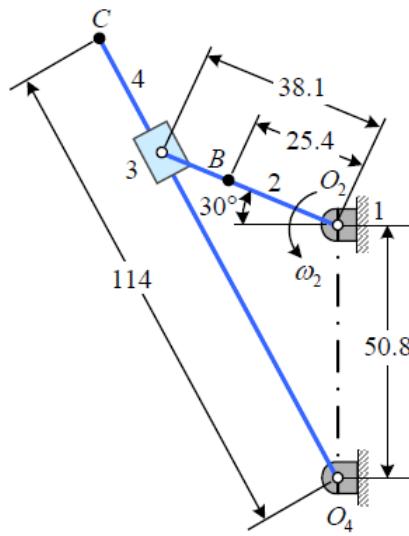
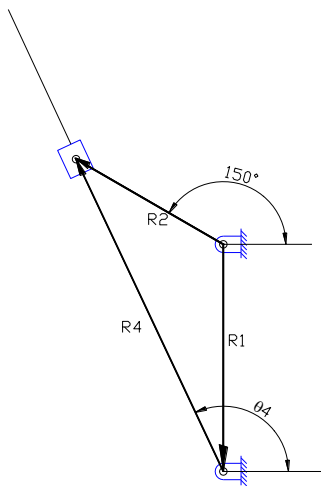


图 5.21

解：



$$150 \text{ r/min} = 150 \times 2\pi / 60 \text{ rad/s} = 15.7 \text{ rad/s}$$

$$R_2 - R_4 - R_1 = 0$$

$$38.1e^{j\theta_2} - r_4e^{j\theta_4} - r_1e^{j\theta_1} = 0$$

$$\text{Re}: 38.1\cos\theta_2 - r_4\cos\theta_4 = 0 \Rightarrow r_4\cos\theta_4 = -33$$

$$\text{Im}: 38.1\sin\theta_2 - r_4\sin\theta_4 + 50.8 = 0 \Rightarrow r_4\sin\theta_4 = 69.85$$

$$\Rightarrow r_4^2 = (-33)^2 + 69.85^2 \Rightarrow r_4 = 77.25$$

$$\text{代入 Re} \Rightarrow \theta_4 = 115^\circ$$

$$j\omega_2 \times 38.1e^{j\theta_2} - \dot{r}_4e^{j\theta_4} - j\omega_4 \times r_4e^{j\theta_4} = 0$$

$$\text{Re}: -38.1\omega_2\sin\theta_2 - \dot{r}_4\cos\theta_4 + \omega_4 \times r_4\sin\theta_4 = 0$$

$$\text{Im}: 38.1\omega_2\cos\theta_2 - \dot{r}_4\sin\theta_4 - \omega_4 \times r_4\cos\theta_4 = 0$$

$$\text{Re} \times \cos\theta_4 + \text{Im} \times \sin\theta_4 \Rightarrow -38.1\omega_2\sin\theta_2\cos\theta_4 + 38.1\omega_2\cos\theta_2\sin\theta_4 - \dot{r}_4 = 0 \Rightarrow \dot{r}_4 = -343 \text{ mm/s}$$

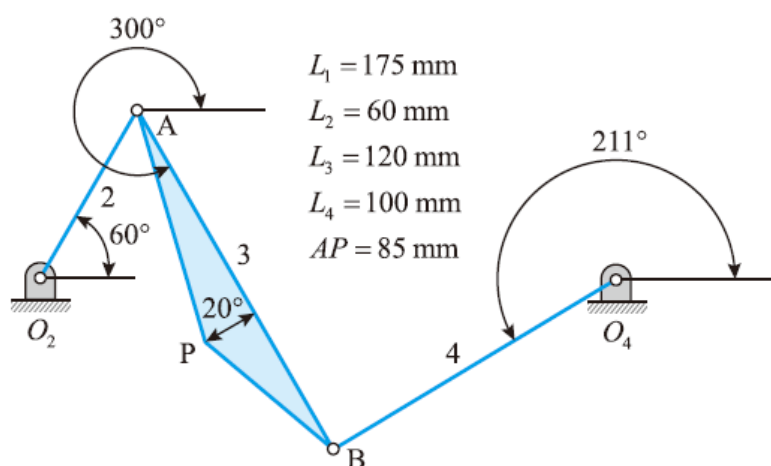
$$\text{代入 Re} \Rightarrow \omega_4 = \frac{38.1\omega_2\sin\theta_2 + \dot{r}_4\cos\theta_4}{r_4\sin\theta_4} = 6.34 \text{ rad/s}$$

$$V_C = O_4C \times \omega_4 = 723 \text{ mm/s} \angle 205^\circ$$

5.11 若圖 5.23 中第二桿之角速度 $\omega_2 = 10 \text{ rad/s}$ 定速旋轉，請以向量迴路法

(複數向量法) 求解圖5.23 第四桿之角速度及角加速度及P 點之速度

及加速度。



5.23

解：

$$R_2 + R_3 - R_4 - R_1 = 0$$

$$60e^{j\theta_2} + 120e^{j\theta_3} - 100e^{j\theta_4} - 175e^{j\theta_1} = 0$$

$$j\omega_2 \times 60e^{j\theta_2} + j\omega_3 \times 120e^{j\theta_3} - j\omega_4 \times 100e^{j\theta_4} = 0$$

$$\text{Re} : -60\omega_2 \sin \theta_2 - 120\omega_3 \sin \theta_3 + 100\omega_4 \sin \theta_4 = 0$$

$$\text{Im} : 60\omega_2 \cos \theta_2 + 120\omega_3 \cos \theta_3 - 100\omega_4 \cos \theta_4 = 0$$

$$\text{Re} \times \cos \theta_3 + \text{Im} \times \sin \theta_3$$

$$-60\omega_2 \sin \theta_2 \cos \theta_3 + 100\omega_4 \sin \theta_4 \cos \theta_3 + 60\omega_2 \cos \theta_2 \sin \theta_3 - 100\omega_4 \cos \theta_4 \sin \theta_3 = 0$$

$$-60\omega_2 \sin(\theta_2 - \theta_3) + 100\omega_4 \sin(\theta_4 - \theta_3) = 0, \quad \theta_2 = 60^\circ, \theta_3 = 300^\circ, \theta_4 = 211^\circ$$

$$\omega_4 = \underline{\underline{-32.65 \text{ rad/s}}}$$

$$\text{Re} \times \cos \theta_4 + \text{Im} \times \sin \theta_4$$

$$-60\omega_2 \sin \theta_2 \cos \theta_4 - 120\omega_3 \sin \theta_3 \cos \theta_4 + 60\omega_2 \cos \theta_2 \sin \theta_4 + 120\omega_3 \cos \theta_3 \sin \theta_4 = 0$$

$$-60\omega_2 \sin(\theta_2 - \theta_4) - 120\omega_3 \sin(\theta_3 - \theta_4) = 0,$$

$$\omega_3 = 15.23 \text{ rad/s}$$

$$-\omega_2^2 \times 60e^{j\theta_2} + j\alpha_2 \times 60e^{j\theta_2} - \omega_3^2 \times 120e^{j\theta_3} + j\alpha_3 \times 120e^{j\theta_3} + \omega_4^2 \times 100e^{j\theta_4} - j\alpha_4 \times 100e^{j\theta_4} = 0$$

$$\text{Re} : -60\omega_2^2 \cos \theta_2 - 60\alpha_2 \sin \theta_2 - 120\omega_3^2 \cos \theta_3 - 120\alpha_3 \sin \theta_3 + 100\omega_4^2 \cos \theta_4 + 100\alpha_4 \sin \theta_4 = 0$$

$$\text{Im} : -60\omega_2^2 \sin \theta_2 + 60\alpha_2 \cos \theta_2 - 120\omega_3^2 \sin \theta_3 + 120\alpha_3 \cos \theta_3 + 100\omega_4^2 \sin \theta_4 - 100\alpha_4 \cos \theta_4 = 0$$

$$\text{Re} : 120\alpha_3 \sin \theta_3 - 100\alpha_4 \sin \theta_4 = -60\omega_2^2 \cos \theta_2 - 60\alpha_2 \sin \theta_2 - 120\omega_3^2 \cos \theta_3 + 100\omega_4^2 \cos \theta_4$$

$$\text{Im} : -120\alpha_3 \cos \theta_3 + 100\alpha_4 \cos \theta_4 = -60\omega_2^2 \sin \theta_2 + 60\alpha_2 \cos \theta_2 - 120\omega_3^2 \sin \theta_3 + 100\omega_4^2 \sin \theta_4$$

$$\text{Re} \times \cos \theta_3 + \text{Im} \times \sin \theta_3 \rightarrow \alpha_4 = \underline{\underline{229.8 \text{ rad/s}^2}}$$

$$\text{Re} \times \cos \theta_4 + \text{Im} \times \sin \theta_4 \rightarrow \alpha_3 = 1156 \text{ rad/s}^2$$

$$V_p = j\omega_2 \times 60e^{j\theta_2} + j\omega_3 \times 85e^{j(\theta_3 - 20^\circ)} = 601.5 + j947.3 \Rightarrow V_p = \underline{\underline{1122 \text{ mm/s}, \angle 57.6^\circ}}$$

$$A_p = -\omega_2^2 \times 60e^{j\theta_2} + j\alpha_2 \times 60e^{j\theta_2} - \omega_3^2 \times 85e^{j(\theta_3 - 20^\circ)} + j\alpha_3 \times 85e^{j(\theta_3 - 20^\circ)} = 90340 + j31280$$

$$\Rightarrow A_p = \underline{\underline{95600 \text{ mm/s}^2, \angle 19^\circ}}$$